ST. C. F. ANDREWS SCHOOL

CHEMISTRY PROJECT

Session: 2024 – 2025

A Project Report On-:

To Study Oxalate Ion Content in Guava Fruit

Submitted by Nandini Raj[put

Class- 12 Science

Under Guidance of

Mr Rajeev Sharma

(Chemistry Teacher)

Index

Certificate

ST. C. F. ANDREWS SCHOOL

This is to certify that *Nandini Rajput* of Class **12 Science** has successfully completed the Chemistry investigatory project titled:

"To Study Oxalate Ion Content in Guava Fruit"

under the guidance of **Mr. Rajeev Sharma**, Chemistry Teacher, during the academic year **2024-25**.

The project work is genuine and has been carried out with utmost sincerity and dedication by the student.

Subject Teacher: ___________________ **External Examiner**: ___________________

Principal: ___________________

Declaration

I, **Nandini Rajput**, a student of Class **12 Science**, hereby declare that the investigatory project titled **"To Study Oxalate Ion Content in Guava Fruit"** submitted to **St Cf Andrews School** is my original work.

This project was carried out under the guidance and supervision of **Mr. Rajeev Sharma**, Chemistry Teacher, as part of the Chemistry curriculum for the academic year **2024-25**.

I further declare that this project is genuine and has not been submitted by anyone else for any academic purpose. The data and observations presented in this project are based on the experiments conducted by me with due sincerity and diligence.

Student: ____________________
(Nandini Rajput)

Date: ____________________

Acknowledgment

I would like to express my sincere gratitude to everyone who guided and supported me throughout the completion of my investigatory project titled **"To Study Oxalate Ion Content in Guava Fruit"**.

First and foremost, I am deeply grateful to **Mr. Rajiv Sharma**, my Chemistry teacher, for his invaluable guidance, encouragement, and support throughout this project. His expertise and suggestions have been instrumental in the successful completion of this work.

I extend my heartfelt thanks to the **Principal** and the **staff of [School Name]** for providing the necessary resources and facilities to conduct this project.

I am also thankful to my **family and friends** for their constant encouragement and motivation during the entire process. Their support gave me the strength and inspiration to complete this project with dedication.

Lastly, I would like to thank the **CBSE curriculum** for including this practical component, which has enhanced my understanding of the subject through hands-on learning.

Student: ***Nandini Rajput***

Aim

To determine the oxalate ion content in guava fruit at different ripening stages, study the effect of temperature on oxalate ion extraction, and compare the efficiency of mechanical and boiling extraction methods.

Objectives

1. To understand the chemical properties of oxalate ions in guava fruit.
2. To determine the effect of ripening on oxalate ion concentration.
3. To evaluate how temperature influences the extraction of oxalates.
4. To compare the efficiency of two extraction methods: mechanical and boiling.

Introduction

Guava (Psidium guajava) is a common tropical fruit known for its rich nutritional content, particularly Vitamin C, dietary fiber, and minerals. However, guava also contains oxalates, which are organic compounds that can form insoluble salts with calcium, leading to kidney stones when consumed in excess. Oxalates naturally occur in plants and play a role in defense mechanisms. This project investigates the oxalate ion content in guava at different ripening stages and evaluates the efficiency of oxalate ion extraction under varying conditions. This study aims to provide insights into how the chemical composition of guava changes during ripening and how oxalates can be efficiently extracted for analysis.

Apparatus Required

- **Beakers** (250 mL and 100 mL)
- **Conical Flasks** (100 mL)
- **Burette** (50 mL)
- **Pipette** (10 mL)
- **Pestle and Mortar** (for crushing guava)
- **Weighing Balance** (for weighing guava fruit)
- **Filter Paper**
- **Funnels**
- **Distilled Water**
- **Dilute Sulfuric Acid** (H_2SO_4, 1 M)
- **Potassium Permanganate Solution** (0.1 M)
- **Water Bath** (for heating the guava extract)
- **Thermometer** (for measuring the temperature of the extract)
- **Glass Stirring Rod**
- **Graduated Cylinder**
- **Burette Clamp**
- **Wash Bottle** (for rinsing apparatus)
- **Test Tubes** (for small samples of guava extract)

Experiment 1: Oxalate Ion Content in Guava at Different Ripening Stages

Objective:
To determine the oxalate ion content in guava at three ripening stages: unripe, semi-ripe, and ripe.

Procedure:

1. **Preparation of Guava Samples**:
   50 grams of guava fruit was taken from each ripening stage—unripe, semi-ripe, and ripe. The samples were washed and chopped into small pieces. Each sample was crushed into a pulp using a pestle and mortar to release the cellular contents.
2. **Extraction of Oxalate Ions**:
   Fifty milliliters of distilled water was added to the pulp, and the mixture was stirred for several minutes to dissolve oxalates into the water. The solution was then filtered to obtain a clear extract.
3. **Titration of Extract**:
   Ten milliliters of the filtrate was taken into a conical flask, and 10 mL of dilute sulfuric acid (H_2SO_4) was added. The solution was titrated with 0.1M potassium permanganate ($KMnO_4$) until a light pink color persisted. The procedure was repeated for all three stages of guava.

Observations:

Ripening Stage	Volume of $KMnO_4$ Used (mL)
Unripe	18.2
Semi-Ripe	13.4
Ripe	8.7

Conclusion:

The oxalate ion content decreases as the guava ripens, likely due to enzymatic breakdown during ripening.

Experiment 2: Effect of Temperature on Oxalate Ion Extraction

Objective:
To study the effect of temperature on the efficiency of oxalate ion extraction.

Procedure:

1. **Division of Guava Sample**:
   A semi-ripe guava was crushed into a pulp, and the pulp was divided into three equal portions.
2. **Application of Temperature**:
   One portion was left at room temperature (25°C), the second was heated to 40°C using a water bath, and the third was heated to 60°C. Each sample was maintained at the respective temperature for 10 minutes.
3. **Filtration and Titration**:
   The mixtures were filtered, and 10 mL of the filtrate from each sample was taken into separate conical flasks. Ten milliliters of dilute sulfuric acid was added to each solution, which was then titrated with 0.1M $KMnO_4$ until the endpoint was reached.

Observations:

Temperature (°C)	Volume of $KMnO_4$ Used (mL)
25	10.5
40	11.2
60	13.6

Conclusion:

Higher temperatures improve the extraction efficiency of oxalate ions, likely due to increased solubility and faster reaction rates at elevated temperatures.

Experiment 3: Comparison of Extraction Methods

Objective:
To compare the efficiency of mechanical extraction versus boiling extraction for oxalate ions.

Procedure:

1. **Mechanical Extraction**:
   A 50-gram sample of guava was crushed into a pulp, mixed with 50 mL of distilled water, and stirred thoroughly. The mixture was filtered to obtain a clear extract.
2. **Boiling Extraction**:
   Another 50-gram sample of guava was crushed into a pulp and mixed with 50 mL of distilled water. The mixture was boiled for 5 minutes, cooled, and then filtered.
3. **Titration of Extracts**:
   For each method, 10 mL of the filtrate was taken into a conical flask, and 10 mL of dilute sulfuric acid was added. The solutions were titrated with 0.1M $KMnO_4$ until the endpoint was reached.

Observations:

Method	Volume of $KMnO_4$ Used (mL)
Mechanical	9.3
Boiling Extraction	14.8

Conclusion:

Boiling extraction releases more oxalates as it breaks down cell walls, making the method more effective than mechanical extraction.

Precautions

1. Ensure accurate measurement of solutions and volumes during titration.
2. Maintain a clean and dry burette and conical flask for precise results.
3. Handle potassium permanganate carefully, as it is a strong oxidizing agent.
4. Avoid overheating guava samples during temperature-based experiments.

Bibliography

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3. www.scienceexperiments.com
4. Research journals on oxalates in tropical fruits